

Exercise 5

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Exercise 2.6(E.2.6, page 115)

Show that the following orthogonality relation

$$\int \left[\chi_{m,k} \left(\frac{\hbar(k+k')}{2} + eBy \right) \chi_{n,k'} \right] dy = \delta_{k,k'}$$

where the two function $\chi_{m,k}(y)$ and $\chi_{n,k'}(y)$ satisfy

$$\begin{aligned} \left[E_s + \frac{(\hbar k + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y) \right] \chi_{m,k}(y) &= E \chi_{m,k}(y) \\ \left[E_s + \frac{(\hbar k' + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y) \right] \chi_{n,k'}(y) &= E \chi_{n,k'}(y) \end{aligned}$$

Proof. Substituting $\chi_{n,k'}(y)$ and $\chi_{m,k}(y)$ into the Schrödinger equation, we have

$$\begin{aligned} \chi_{n,k'}(y) \left[E_s + \frac{(\hbar k + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y) \right] \chi_{m,k}(y) &= E \chi_{m,k}(y) \\ \chi_{m,k}(y) \left[E_s + \frac{(\hbar k' + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y) \right] \chi_{n,k'}(y) &= E \chi_{n,k'}(y) \end{aligned}$$

Subtracting the two equations, we get

$$\begin{aligned} &\chi_{n,k'}(y) \left[E_s + \frac{(\hbar k + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y) \right] \chi_{m,k}(y) \\ &- \chi_{m,k}(y) \left[E_s + \frac{(\hbar k' + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y) \right] \chi_{n,k'}(y) \\ &= E \chi_{m,k}(y) \chi_{n,k'}(y) - E \chi_{n,k'}(y) \chi_{m,k}(y) \\ &\Rightarrow \chi_{n,k'}(y) \frac{(\hbar k + eBy)^2 - (\hbar k' + eBy)^2}{2m} \chi_{m,k}(y) = 0 \\ &\Rightarrow \chi_{n,k'}(y) \frac{\hbar(k - k')(\hbar(k + k') + 2eBy)}{2m} \chi_{m,k}(y) = 0 \end{aligned}$$

Integrating both sides over y , we have

$$\int \chi_{n,k'}(y) \frac{\hbar(k + k') + 2eBy}{2m} \chi_{m,k}(y) dy = 0, \quad \text{if } k \neq k'$$

By normalizing the wave function in appropriate way, we have

$$\int \chi_{m,k} \left(\frac{\hbar(k+k')}{2} + eBy \right) \chi_{n,k'} dy = \delta_{k,k'}$$

□